Table of Contents

iMotions Lab > Analysis > Signal Processing (R Notebooks) > Facial Expression Analysis > AFFDEX

R Notebooks (FEA): AFFDEX Raw Data Statistics Export 2

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R Notebooks (FEA): AFFDEX Raw Data Statistics Export

The R Notebook is called: AFFDEX Raw Data Statistics Export. It produces file export for AFFDEX Statistics on raw data. The notebook calculates parametric (mean, var, sd), robust (quantiles, IQR) and moment (moment, skewness, kurtoses) statistics for chosen stimuli, scenes, annotations and/or fragments. Please note that you can directly go export data from iMotions UI, without adding an additional R Notebook in the Signal Processing section. This will be explained more later in How to Export in iMotions Desktop.

This article includes:

1. Prerequisites

2. How to Export in iMotions Desktop

3. Outputs: Exported files and some basics of each statistics

1. Prerequisites

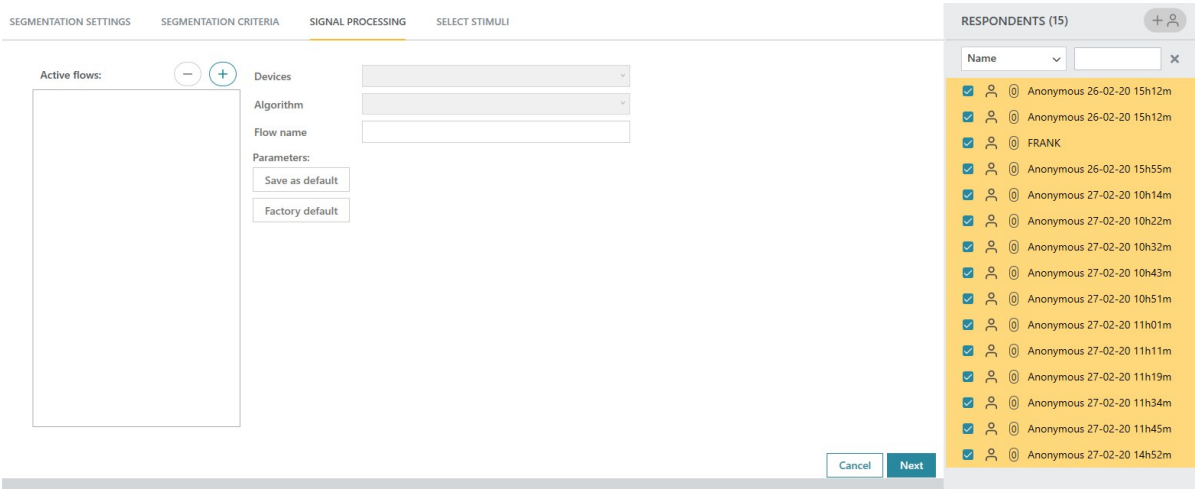
Supported iMotions version: iMotions 9.2.1 and later

Supported signals: AFFDEX

2. How to Export in iMotions Desktop

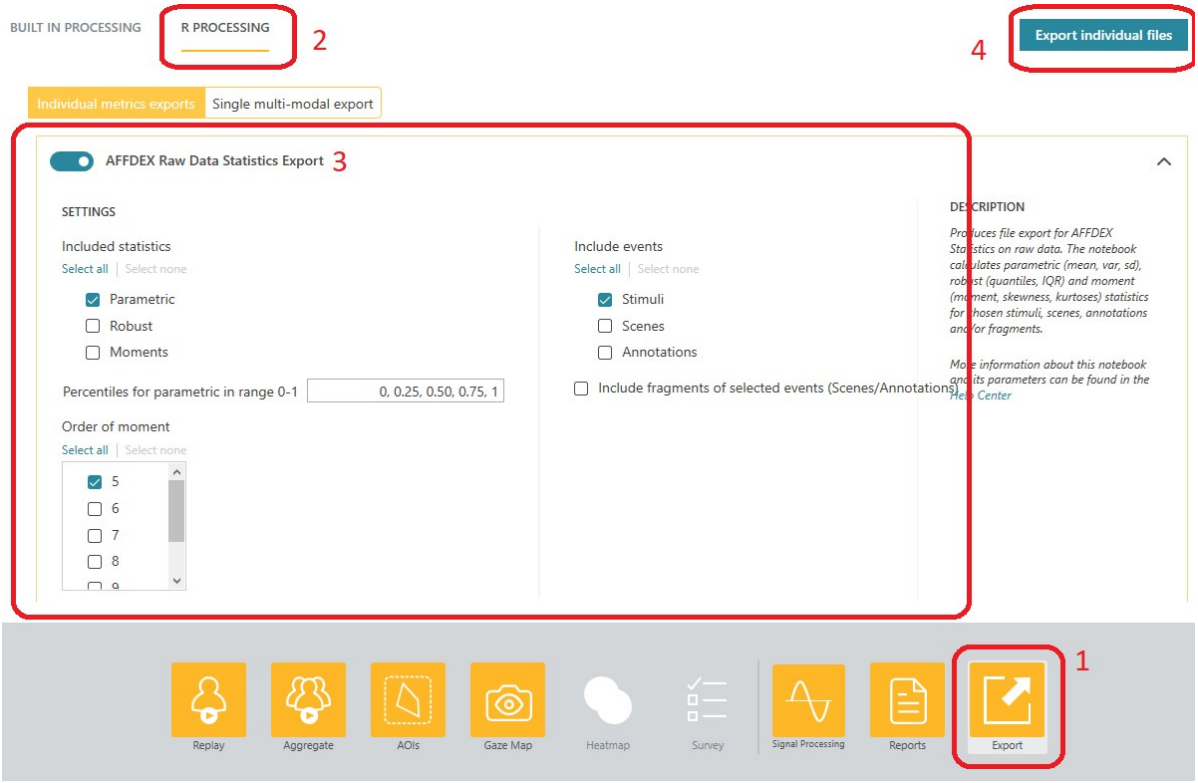
Create your analysis that includes your respondents and stimuli of interest. When creating your Analysis under your Study, you do NOT need to add any additional R Notebook under the "Signal Processing" section if you only want to export AFFDEX statistics.

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Leaving the "Active flows" blank is fine if you only need to export AFFDEX statistics.

Then proceed to Export button from the bottom menu when in the analysis level. You find the export selection for the "Affdex Raw Data Statistics Export" under the tab "R PROCESSING", and can then choose the export settings you need. Export by clicking "Export" in the upper right corner.



Export Settings Explained:

Included statistics: Choose one or more types of statistics to be included in your export file.

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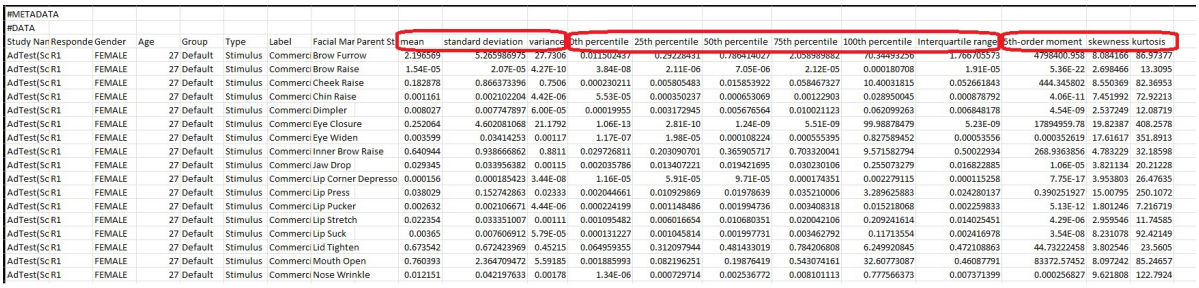
Percentiles for parametric in range 0-1: The wanted percentiles to be calculated for statistics "quantiles".

Order of moment: The order of moment wanted to be calculated for statistics "moment". You can choose multiple moments (above or equal to order 5) at once for the export.

Include events: choose stimulus types you would like to calculate the statistics from - Stimulus, Scene, Annotations, and with or without fragments.

3. Outputs: Exported files and basics of each statistics

The exported file with everything included will look like this:



#METADATA
#DATA

Study Nan	Responde	Gender	Age	Group	Type	Label	Facial Mar	Parent St	mean	standard deviation	variance	0th percentile	25th percentile	50th percentile	75th percentile	100th percentile	Interquartile range	5th-order moment	skewness	kurtosis
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Brow Furrow		2.196569	5.265986975	27.7306	0.011502437	0.29228431	0.786414027	2.058989882	70.34493256	1.766705573	4798400.958	8.084166	86.97377
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Brow Raise		1.54E-05	2.07E-05	4.27E-10	3.84E-08	2.11E-06	7.05E-06	2.12E-05	0.000180708	1.91E-05	5.36E-22	2.698466	13.3095
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Cheek Raise		0.182878	0.866373396	0.7506	0.000230211	0.005805483	0.015853922	0.058467327	10.40031815	0.052661843	444.345802	8.550369	82.36953
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Chin Raise		0.001161	0.002102204	4.42E-06	5.53E-05	0.000350237	0.000653069	0.00122903	0.028950045	0.000878792	4.06E-11	7.451992	72.92213
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Dimpler		0.008027	0.007747897	6.00E-05	0.00019955	0.003172945	0.005676564	0.010021123	0.062099263	0.006848178	4.54E-09	2.537249	12.08719
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Eye Closure		0.252064	4.602081068	21.1792	1.06E-13	2.81E-10	1.24E-09	5.51E-09	99.98878479	5.23E-09	17894959.78	19.82387	408.2578
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Eye Widen		0.003599	0.03414253	0.00117	1.17E-07	1.98E-05	0.000108224	0.000555395	0.827589452	0.00053556	0.000352619	17.61617	351.8913
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Inner Brow Raise		0.640944	0.938666862	0.8811	0.029726811	0.203090701	0.365905717	0.703320041	9.571582794	0.50022934	268.9363856	4.783229	32.18598
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Jaw Drop		0.029345	0.033956382	0.00115	0.002035786	0.013407221	0.019421695	0.030230106	0.255073279	0.016822885	1.06E-05	3.821134	20.21228
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Lip Corner Depresso		0.000156	0.000185423	3.44E-08	1.16E-05	5.91E-05	9.71E-05	0.000174351	0.002279115	0.000115258	7.75E-17	3.953803	26.47635
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Lip Press		0.038029	0.152742863	0.02333	0.002044661	0.010929869	0.01978639	0.035210006	3.289625883	0.024280137	0.390251927	15.00795	250.1072
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Lip Pucker		0.002632	0.002106671	4.44E-06	0.000224199	0.001148486	0.001994736	0.003408318	0.015218068	0.002259833	5.13E-12	1.801246	7.216719
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Lip Stretch		0.022354	0.033351007	0.00111	0.001095482	0.006016654	0.010680351	0.020042106	0.209241614	0.014025451	4.29E-06	2.959546	11.74585
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Lip Suck		0.00365	0.007606912	5.79E-05	0.000131227	0.001045814	0.001997731	0.003462792	0.11713554	0.002416978	3.54E-08	8.231078	92.42149
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Lid Tighten		0.673542	0.672423969	0.45215	0.064959355	0.312097944	0.481433019	0.784206808	6.249920845	0.472108863	44.73222458	3.802546	23.5605
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Mouth Open		0.760393	2.364709472	5.59185	0.001885993	0.082196251	0.19876419	0.543074161	32.60773087	0.46087791	83372.57452	8.097242	85.24657
AdTest(Sc	R1	FEMALE	27	Default	Stimulus	Commerci	Nose Wrinkle		0.012151	0.042197633	0.00178	1.34E-06	0.000729714	0.002536772	0.008101113	0.777566373	0.007371399	0.000256827	9.621808	122.7924

More about the statistics*:*

It is highly recommended that you are fully aware of the mechanism behind the statistics that you are planning to use before you apply them in your research analysis.

Parametric Statistics:

mean: The arithmetic mean, also known as arithmetic average, is a central value of a finite set of numbers - here the probability scores of each facial marker within that chosen stimulus duration. *(resources)*

standard deviation: *The standard deviation is a measure of the amount of variation or dispersion of a set of values - here the probability scores of each facial marker within that chosen stimulus duration. A low standard deviation indicates that the values tend to be close to the mean of the set, while a high standard deviation indicates that the values are spread out over a wider range. (resources)*

variance: Variance is a measure of dispersion, meaning it is a measure of how far a set of numbers - here the probability scores of each facial marker within that chosen stimulus duration, spread out from their average value. *(resources)*

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Robust Statistics:

nth percentile: Percentile is a number where a certain percentage of scores fall below that number. For example, the 50th percentile is the score of that facial marker at or below which 50% of the scores in the distribution may be found. Percentiles are expressed in the same unit of measurement as the input scores (here the probability scores of each facial marker). One example: you might observe during one stimulus recording, that one respondent scored 40 at one point, for Joy. But that figure has no real meaning when you want to compare and see if this is a high or low score for this respondent during this whole recording - unless you know what percentile it falls into - If you know that this 40 score is in the 75th percentile, that means this 40 scored better than 75% of all other data points for this stimulus recording. *(resources)*

Interquartile range: The interquartile range (IQR) is a measure of statistical dispersion. It is the spread of the dataset - here the probability scores of each facial marker within that chosen stimulus duration. It is defined as the spread difference between the 75th and 25th percentiles of the dataset. The primary use of the IQR is to represent the spread difference between the upper and lower quartiles of a data set. This can be used as an indicator of the variability of the dataset. *(resources)*

Moment Statistics:

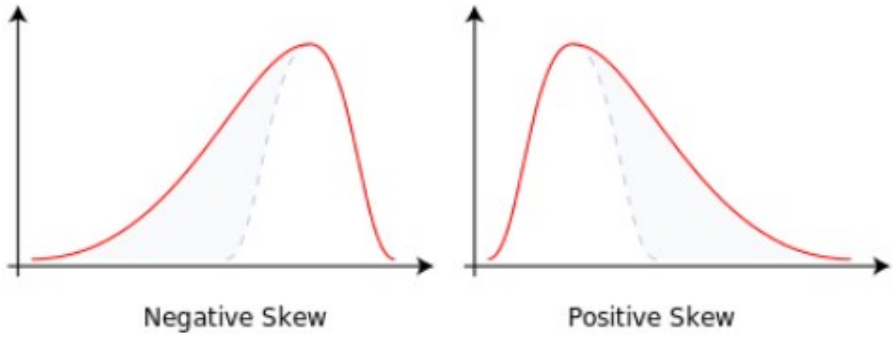
nth-order moment: In mathematics, the moments of a probability distribution are quantitative measures related to the shape of the probability distribution.

Moments below 4th-order moments: The first moment is the mean; The second moment is the variance; The third moment is skewness; The fourth moment is kurtosis.

Higher-order moments involve non-linear combinations of the data and can be used for the description or estimation of further shape parameters. The higher the moment, the harder it is to estimate, in the sense that larger samples are required in order to obtain estimates of similar quality. Please make sure you are fully informed of the relevant statistics knowledge before you use it. *(resources)*

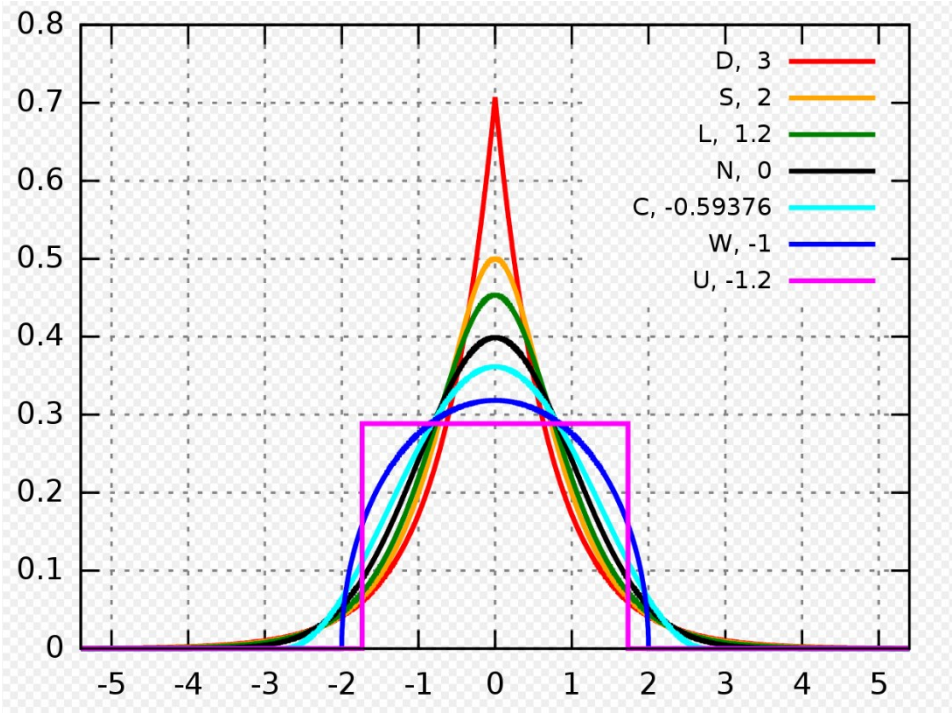
skewness: skewness is the measure of the lopsidedness of the distribution: A distribution that is skewed to the left (the tail of the distribution is longer on the left) will have negative skewness. A distribution that is skewed to the right (the tail of the distribution is longer on the right), will have positive skewness. *(resources)*

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negative skew: The left tail is longer; the mass of the distribution is concentrated on the right of the figure. *positive skew*: The right tail is longer; the mass of the distribution is concentrated on the left of the figure. (resource from *Wikipedia*)

kurtosis: kurtosis is a measure of the heaviness of the tail of the distribution, compared to the normal distribution of the same variance. E.g., for a given amount of dispersion, higher kurtosis corresponds to thicker tails, while lower kurtosis corresponds to broader shoulders. *(resources)*



Probability density functions for selected distributions with mean 0, variance 1 and different excess kurtosis: one can see from the plot, the higher kurtosis corresponds to thicker tails (red line), while lower kurtosis corresponds to broader shoulders (pink line).
(resource from *Wikipedia*)

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